

5. Experiments

5.1. Simulation Setup

We evaluate our TradingAgents framework using a comprehensive backtesting simulation from January 1st to March 29th, 2024, across major technology stocks including Apple, Nvidia, Microsoft, Meta, and Google. TradingAgents facilitates seamless plug-and-play strategies during the simulation, enabling straightforward comparisons with any baseline. Agents make decisions based solely on data available up to each trading day, ensuring no future data is used (eliminating look-ahead bias). Based on their analysis, TradingAgents generates trading signals to buy, sell, or hold assets, which are then executed. Afterward, analysis metrics are calculated before proceeding to the next day’s data.

We benchmark against five established strategies: Buy and Hold, MACD, KDJ+RSI, ZMR, and SMA (baseline descriptions in Appendix S1.1). Performance is evaluated using four key metrics: Cumulative Return (CR), Annualized Return (AR), Sharpe Ratio (SR), and Maximum Drawdown (MDD) (formulations in Appendix S1.2).

5.2. Back Trading

To simulate a realistic trading environment, we utilize a multi-asset and multi-modal financial dataset comprising of various stocks such as Apple, Nvidia, Microsoft, Meta, Google, and more. Our multi-modal dataset integrates historical stock prices, news articles, social media sentiment, insider transactions, financial statements, and 60 technical indicators per asset. The dataset includes:

Historical Stock Prices: Open, high, low, close, volume, and adjusted close prices from January 1st, 2024, to March 29th, 2024.

News Articles: Daily news updates are gathered from diverse sources such as Bloomberg, Yahoo, EODHD, FinnHub, and Reddit, covering specific company developments, global events, macroeconomic trends, and government updates.

Social Media Posts and Sentiment: Posts from Reddit, X/Twitter, and other platforms along with sentiment scores of posts calculated by auxiliary language models.

Insider Sentiments and Transactions: Sentiment derived from public information, including transactions from SEDI and relevant company filings.

Financial Statements and Earnings Reports: Quarterly and annual reports filed by companies.

Company Profiles and Financial History: Descriptions of company profiles, target industries, and financial history reported by third parties.

Technical Indicators: Sixty standard technical analysis indicators calculated for each asset, including MACD, RSI, Bollinger Bands, etc.

6. Results and Analysis

In this section, we present the experimental results and analyze the performance of TradingAgents in comparison to the baseline models.

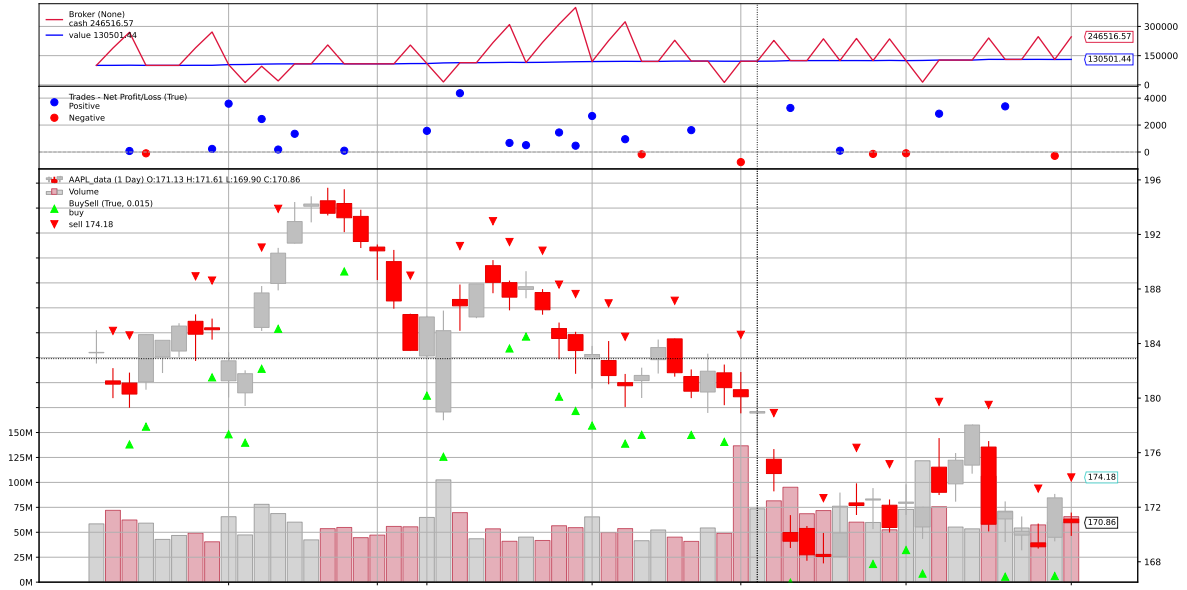


Fig. 6: **TradingAgents** Detailed Transaction History for \$AAPL. Green / Red Arrows indicate Long / Short Positions respectively, showing the model’s trading decisions over time.

6.1. Performance Comparison

6.1.1. Cumulative and Annual Returns

Table 1 and Figures 6, 7, S1, S2, S3, and S4 demonstrate that our method outperforms existing rule-based trading baselines, particularly in profitability, as measured by returns. TradingAgents achieves at least a 23.21% cumulative return and a 24.90% annual return on the three sampled stocks, surpassing the best-performing baselines by a margin of 6.1%. Notably, on \$AAPL stock—a particularly challenging case due to market volatility during the testing period—traditional methods struggled, as their patterns failed to generalize to this situation. In contrast, TradingAgents excelled under these adverse conditions, achieving returns exceeding 26% within months.

Table 1: Performance comparison across all methods using four evaluation metrics. Results highlighted in green represent the best-performing statistic for each model. The improvement row illustrates TradingAgents’s performance gains over the top-performing baselines.

Categories	Models	AAPL				GOOGL				AMZN			
		CR%↑	ARR%↑	SR↑	MDD%↓	CR%↑	ARR%↑	SR↑	MDD%↓	CR%↑	ARR%↑	SR↑	MDD%↓
Market	B&H	-5.23	-5.09	-1.29	11.90	7.78	8.09	1.35	13.04	17.1	17.6	3.53	3.80
Rule-based	MACD	-1.49	-1.48	-0.81	4.53	6.20	6.26	2.31	1.22	-	-	-	-
	KDJ&RSI	2.05	2.07	1.64	1.09	0.4	0.4	0.02	1.58	-0.77	-0.76	-2.25	1.08
	ZMR	0.57	0.57	0.17	0.86	-0.58	0.58	2.12	2.34	-0.77	-0.77	-2.45	0.82
	SMA	-3.2	-2.97	-1.72	3.67	6.23	6.43	2.12	2.34	11.01	11.6	2.22	3.97
Ours	TradingAgents	26.62	30.5	8.21	0.91	24.36	27.58	6.39	1.69	23.21	24.90	5.60	2.11
Improvement(%)		24.57	28.43	6.57	-	16.58	19.49	4.26	-	6.10	7.30	2.07	-

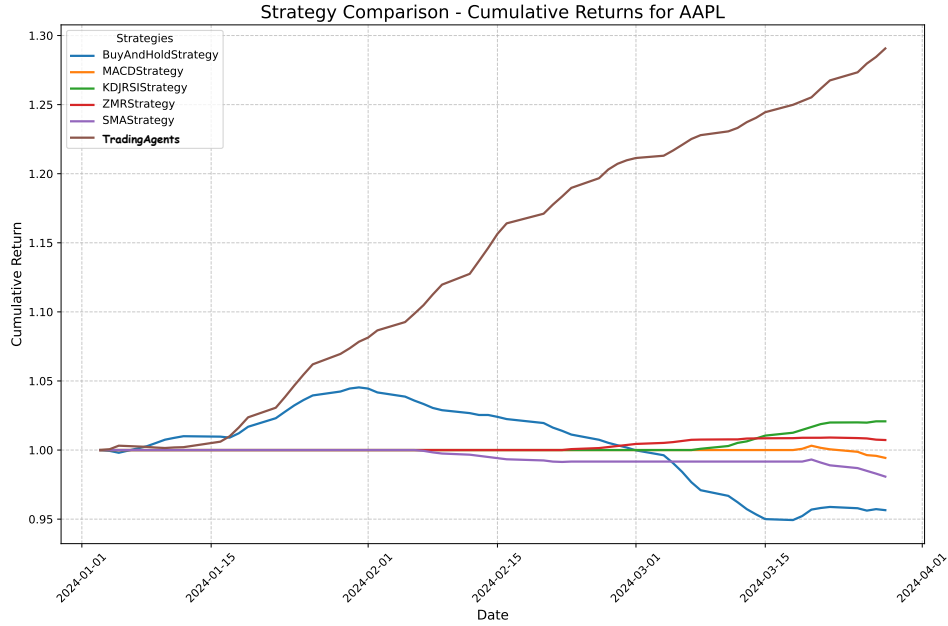


Fig. 7: Cumulative Returns on \$AAPL using **TradingAgents**. The figure shows the performance comparison of our model against baseline approaches for Apple Inc. stock analysis.

6.1.2. Sharpe Ratio

The Sharpe Ratio[†] performance highlights TradingAgents’s exceptional ability to deliver superior risk-adjusted returns, surpassing all baseline models. This result underscores TradingAgents’s effectiveness in balancing returns and risk—a crucial factor for sustainable and predictable investment growth. TradingAgents outperforms market benchmarks such as Buy-and-Hold and rule-based strategies consistently, demonstrating its adaptability. Its capability to maximize returns while maintaining controlled risk exposure establishes a strong foundation for multi-agent and debate-based automated trading algorithms.

6.1.3. Maximum Drawdown

While rule-based baselines demonstrated superior performance in controlling risk, as reflected by their maximum drawdown scores, they fell short in capturing high returns. This trade-off between risk and reward underscores TradingAgents’s strength as a balanced approach. Despite higher returns being typically associated with higher risks, TradingAgents maintained a relatively low maximum drawdown compared to many baselines. Its effective risk-control mechanisms, facilitated by the debates among risk-control agents, ensured that the maximum drawdown remained within a

[†]We benchmarked TradingAgents over 3 months due to intensive LLM and tool use (11 LLM calls & 20+ tool calls/prediction). The highest Sharpe Ratio exceeds our expected empirical range (SR above 2 – very good, above 3 – excellent). We exported TradingAgents’s decision sequences and examined them to ensure calculation correctness. We believe the exceptionally high SR resulted from the phenomenon that there were few pullbacks in TradingAgents during that period. We report results as they are in our experiments faithfully. Future work will optimize LLM reasoning & tool use to enable longer backtesting under limited budgets.